02464 Lecture Notes

Motor Control

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1. Introduction

Why do humans and other animals need a brain? Neuroscientists[1] believe that the only reason we need a brain is to produce complex and adaptable movements. Moving is our only way to interact with and affect the world around us, through movement of arms and legs, speech, gestures, writing, etc. While other cognitive processes such as perception and memory are all important, they are arguably important because they drive or suppress future movements. Evidence for this comes from the sea squirt, an animal that has a nervous system in its adolescence, during which it moves around, exploring its environment. At some point it implants on a rock, and once it has done so, it digests its own brain and nervous system for food. Therefore, once it no longer needs to move, it no longer needs a brain.

Why is this relevant for AI? If we take chess as an example, and simplify it into two tasks. First, how do we decide which piece to move where? When pinning Garry Kasparov against an AI in a game of chess – in this case, IBM's Deep Blue – the AI will win some of the time, against arguably one of the best players of all time. This is thus a solvable problem for the AI, which requires a simple algorithm. It has to look at all the possible moves to the end of the game, and play the ones that will lead to its win.

But if we compare how humans (even small children) physically move chess pieces around to the world's best robots, manipulating them and moving them from one place to another, the robots do not stand the slightest chance. This is because motor control is very complicated, and we still have a poor understanding of how the human (or animal) brain

[1]They call themselves motor chauvinists. Source: Daniel Wolpert

controls movements. The reason for this is because there is much noise and uncertainty in motor control. Our sensory feedback is noisy, our motor system is noisy, and the environment which we interact with is noisy. There are also many motor commands to choose from. And yet, we do not put much apparent conscious effort into deciding how to move when it comes to basic tasks. We walk, talk, and manipulate objects mostly with ease, and yet, generating the coordinated movement that enables us to do these tasks is extremely complex. So the tasks that are very difficult for robotic systems to achieve are routine for us.

So how does the brain decide which motor commands to send to our muscles?

The field of motor control formulates underlying principles that explain how we move our bodies, the decisions we make to do so, etc. Therefore, it formulates mathematical models that capture the mechanics of the body or task, describe how motor commands are selected, and explain the role prior experience has in determining our future behaviour.

In this chapter, we will discuss simplified models and principles of movement neuroscience. These are not only important for the design of robots and embodied AI, but also for understanding general mechanisms of the central nervous system. If we can understand the computational principles of how we move, we can also better understand the mechanisms that underlie our understanding of goals, integration of sensory and memory processes, and even complex social interactions (e.g. we interact successfully because we successfully predict other people's actions, and form beliefs about their goals and intentions from these actions; but we will get to this later in the course).

2. Internal models

Motor control is concerned with the relationship between motor commands and sensory signals, and models that implement the transformation from motor commands to sensory signals, or vice versa, are known as internal models. The central nervous system sends motor commands to our muscles in order to activate or inhibit them. For example, if you want to open a door, you need to know the location of the door handle, as well as the position of your arm.

Simultaneously, the CNS predicts the sensory consequences of the motor command. For example, what are the sensory consequences of opening the door? Moving the door handle will produce a sound (auditory consequence), and you will also see the hand move (visual consequence), and feel the impact of your hand on the door handle (tactile consequence). Such sensory consequences can be predicted based on previous experience, and they are also updated based on new experiences.

The transformation from motor commands to sensory consequences is modeled using "forward internal models". They model the causal relationship between actions and sensory consequences. The "forward" aspect implies that they are predictors – hence, their function is to predict the behaviour of the body as well as the world.

The transformation from desired sensory consequences to motor commands (actions) is modeled using "inverse internal models".

Fortunately for motor control, the CNS does not represent all possible motor activations, otherwise it would be faced with the "curse of dimensionality". The human body has about 600 muscles, and if we consider that each one can be contracted or relaxed, this would lead to 2^{600} possible motor activations to choose from – which is more than the number of atoms in the universe! Thus, a compact representation, known as the "state" of the system, is extracted. For example, if we want to move our hand to a particular location, the state of the hand can be described as the set of activations of the relevant groups of muscles, as well as the position and velocity of the hand.

In addition to the state, the CNS also needs to represent the "context" of the movement, which represents the slowly changing parameters. For example, if we want to lift an object, the context would contain information about the identity of that object, and the mass of our arm and hand.

The sensorimotor loop thus contains 3 models:

1. Given a particular task and the relevant state, the motor command is generated via the inverse model.
2. Given the motor command, the CNS predicts how the state changes (relative to the previous state), using a forward dynamic model.
3. Given this new state and motor command, the CNS then predicts the sensory feedback, using a forward sensory model.

2.1. Forward Models

Consider the task of catching a ball. The ball is moving fast, and if we use the retinal location to estimate its position, there is a delay in our estimate by about 100 ms. Moreover, we cannot directly observe the ball's spin. Rather, we can estimate the spin using sensory information, in this case the ball's path, that we integrate over time, but this will be a noisy estimate. Simultaneously, we need to have an estimate of our arm's current state. The CNS is thus faced with delays in the transduction and transport of sensory signals, as well as noise in sensory and motor signals.

In order to circumvent these challenges, given that we have to act fast to catch the ball, our CNS uses a forward dynamic model to predict the new state of our arm (with an input of our current state), that is, where the arm is going next. Similarly, the CNS also uses a forward model of the external environment, predicting the ball's next state – where the ball is going to go based on where it is now. This estimate is improved by knowing how the ball was thrown – i.e. the motor command – together with a forward model of the ball's dynamics.

The purpose of forward models is to compensate for sensorimotor delays and reduce the uncertainty of the estimates of the state (e.g. of the ball or our hand).

Furthermore, the CNS uses forward models to estimate the sensory outcome of our own actions (referred to as the sensory consequences), before the sensory feedback is available (Fig. 1). For example, if we reach for a glass of water, we predict how the hand will move in the planned direction, or how the glass will feel against our hand (is the water cold or warm?). Or, if we want to speak up, we predict to hear the sound of our voice. The forward model is used to anticipate the sensory signals. These signals have two causes: those that result from self-generated movement, and those that result from the external environment.

In addition to the motor command, the motor system creates a copy of the motor command, known as the efference copy, to generate predictions of the sensory consequences of our own actions. The estimated sensory consequences are then compared with the real sensory feedback (Fig. 1). If the feedback matches our prediction, the cortical response is attenuated. If it does not match our prediction, the cortical response is enhanced by the sensory discrepancy.

This is an important mechanism of self-monitoring. The model allows the system to attenuate perceptions of predicted sensations, and amplify the salience of externally caused ones. This is important because it helps us distinguish between signals from our own movements and those from the external world, allowing us to monitor changes in the environment. For example, experiments have shown that when people stroke their own hand with their other hand, it feels less intense, than when their hand is stroked externally by a motor. This is because self-generated forces are perceptually attenuated, as the estimated sensory feedback matches the real sensory feedback.

There is much evidence for existence of forward models in the human brain (and across species). For example, when we move our eyes to redirect our gaze, there is a very large shift in our retinal image, but what we perceive is a static visual image. This has been attributed to the efference copy, which is used by the forward model to predict the retinal consequences of an eye movement and cancel out the predictable self-generated motor input, providing a mechanism for image stabilization.

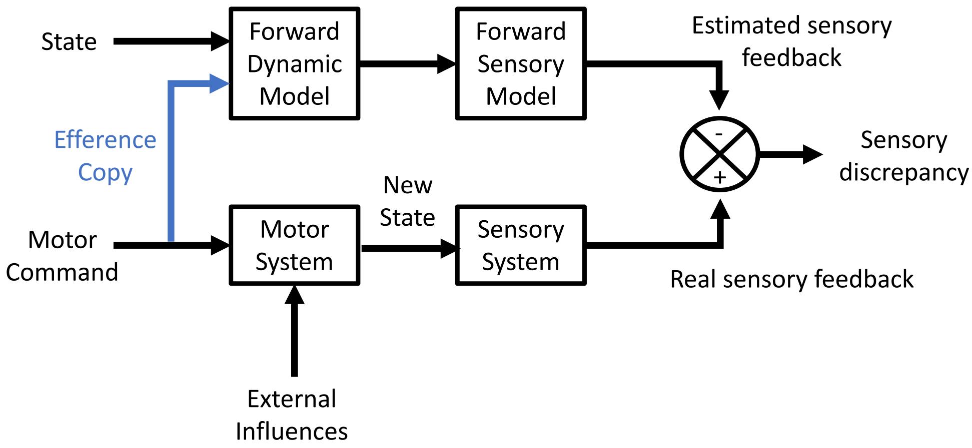

Figure 1: Forward models of motor control (figure adapted from [3]). The forward dynamic model receives an estimate of the current state and a copy of the motor command (efference copy), to generate a prediction of the estimated next state. The forward sensory model then uses this to predict the estimated sensory feedback. The estimated sensory feedback is then compared to the real sensory feedback from the sensory system.

Interestingly, one long standing proposal of what goes wrong in patients with schizophrenia is that they are not able to predict their own action consequences. Therefore, their own actions may feel as if they are caused by external forces. These prediction failures correlate with the severity of delusions of control (i.e. false perceptions and beliefs concerning the intentions of others).

To sum up, forward models are used for:

- estimating future states of the external environment and our own body, because sensory feedback is slow and noisy
- predicting sensory consequences of actions, which is crucial for self-monitoring as they provide a mechanism for distinguishing external influences from self-produced effects (this also has a role in increasing sensitivity to unexpected effects)

2.2. Inverse Models

Inverse models are used to predict the motor command required for desired sensory consequences. It is important to note that both forward and inverse models change throughout life. They change on short time-scales (during real-time interaction with the world), as well as long ones (as we grow and develop).

Learning an inverse model is not an easy task. For example, if we are throwing darts and fail to hit the bulls-eye, we do not have access to which muscle activations should be changed in order to do better. We instead receive errors in the form of sensory signals – e.g., the dart did not hit the bulls-eye, but was close – and then need to convert these to

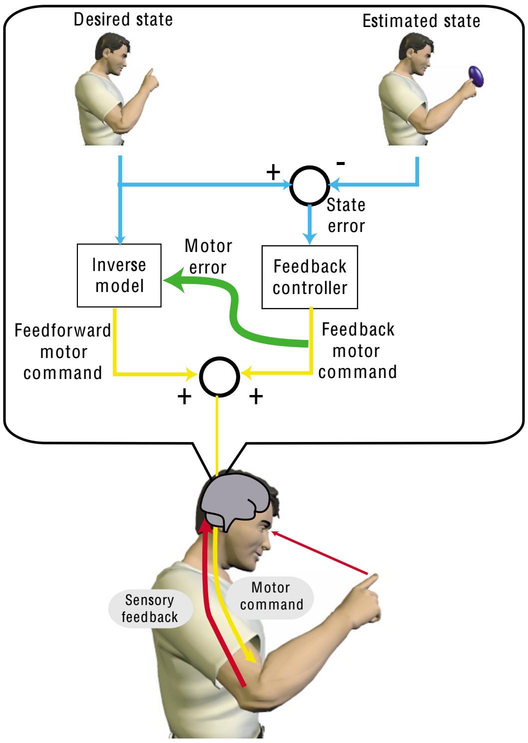

Figure 2: Feedback-error learning model (figure from [4]). We learn an inverse model by training it using the motor error, computed by the feedback controller. The motor error is based on the difference between the desired and estimated state.

motor errors. Then these motor errors can be used to train our inverse model.

One solution is the feedback-error learning model, which computes a motor command based on the estimated state error (difference between the desired state and estimated state). As seen in Fig. 2, the motor command is the sum of the feedforward motor command (output of the inverse model) and the feedback motor command (output of the feedback controller). If there is no difference in the desired state and estimated state, then the feedback controller outputs no motor error (and hence, no feedback motor command), which means that feedforward motor command estimated with the inverse model was accurate. If there is a discrepancy between the estimated and desired state, then the motor error is used to train the inverse model.

Inverse models have been supported by neurophysiological evidence from the cerebellum (part of the brain that is involved in the control of movements). For example, studies have shown that the ocular following response (a reflex eye movement) is predicted based on an inverse model of the eye's dynamics in the cerebellum.

3. Bayes rule in motor control

Now that we have covered the sensorimotor loop, let's consider how we estimate the state of the environment and our body (as well as the context of our actions, but let's leave this for the time being). The internal models we discussed above (forward and inverse models) are embedded in a probabilistic framework [2]. When we interact with the world through movement, we need to integrate our sensory (i.e. perceptual) information together with our motor plans. Hence, there is a tight coupling between action and perception. You have already learned about Bayesian inference in perception, so I will just provide a very brief recap to Bayes rule, and how it relates to action.

Essentially motor control is about mapping sensory inputs to motor outputs. Sensory inputs constitute our "data". We generate our beliefs about the state of the world (and our bodies) in a probabilistic way, based on both the sensory inputs, as well as our prior knowledge (i.e. memory). These beliefs then generate the motor outputs, hence driving our actions.

For example, let's consider once again the task of catching a ball that our friend has thrown to us. We need to make an optimal estimate of the final location of the fast moving ball, where we can catch it. We already know that our sensory information is noisy - i.e. our vision does not provide perfect information about the position or velocity of the ball - hence our estimate will be uncertain. Let's denote our prior belief regarding where the ball will end up as $P(state)$. This is our prediction before we have received any sensory feedback. These priors are typically trained through our experience with the world, i.e., from acquiring experience through playing ball, or playing ball with this particular friend. Maybe we know that when our friend throws the ball, it often skews a little to the left in relation to us, which will contribute to our prior belief. When we receive sensory feedback (see the ball in the air), then our belief is updated. The sensory feedback will then be used to calculate the likelihood, $P(sensory\ input|state)$. This is the probability that we receive the given sensory input, given the current state of the ball (the state of the ball on this particular throw). But since you have some experience with throwing and catching this ball, as well as is playing catch with this particular friend, then according to Bayes rule, your uncertainty in your final sensory estimate is reduced because you have a prior (information not available in the current throw) over all possible ball locations.

We can now work out the probability distribution of the ball's final location in space, by combing the sensory input and the prior knowledge, which would constitute the posterior:

$$P(state|sensory\ input) = \frac{P(sensory\ input|state)P(state)}{P(sensory\ input)}$$

The maximum of the posterior is the most probable location where the ball will go/land.

The denominator, $P(sensory\ input)$, is the normalization constant, so that the sum of probabilities over all the possible states (i.e. all possible final locations of the ball) is one. This can be quite difficult to compute, as it it can be quite high-dimensional. For a continuous number of states, the normalization constant would then be:

$$P(sensory\ input) = \int P(sensory\ input|state)P(state)\ dstate$$

4. Decision theory

The question that remains is what do we do with this posterior? Bayes rule does not specify how we use our current belief to choose a particular action. The choice of each action is driven by the aim to minimize the expected loss (or cost) [1]. (Alternatively, the aim is to maximize the expected reward or utility.) This is specified by quantifying the value of each possible action, given each possible state of the environment, by using a loss function, $L(action, state)$. We quantify the expected loss for a given action as:

$$\sum_{states} L(action, state)P(state|sensory\ input)$$

The loss for each action is thus the sum (or average) of the losses of all possible states for each action, weighted by the posterior of each state. Note that in cases where the loss is actually a reward, the output of the formula would be negative. The action with the lowest expected loss is then chosen.

References

[1] K. P. Körding and D. M. Wolpert. Bayesian decision theory in sensorimotor control. *Trends in Cognitive Sciences*, 10(7):319–326, 2006. Special issue: Probabilistic models of cognition.

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